1. Which part of speech names a person, place, thing, or abstract idea?

(A) Adjective
(B) Noun
(C) Pronoun
(D) Conjunction

2. In the phrase 'purified Taq polymerase', the word 'purified' functions as:

(A) A participle adjective
(B) An adverb
(C) A finite verb
(D) A proper noun

3. Which of the following is an uncountable (mass) noun in scientific English?

(A) Plasmid
(B) Bacterium
(C) Apparatus
(D) Micropipette

4. Which pronoun replaces the subject 'The molecular biologists' in subjective case?

(A) Them
(B) They
(C) Their
(D) Theirs

5. Identify the relative pronoun in: 'The strain that expressed GFP fluoresced.'

(A) The
(B) expressed
(C) that
(D) fluoresced

6. What is the primary function of an adverb in scientific syntax?

(A) Modify nouns
(B) Connect clauses
(C) Modify verbs/adjectives
(D) Express emotion

7. In 'The mixture incubated overnight', 'overnight' functions as:

(A) An adverb of time
(B) An adjective
(C) A preposition
(D) A noun direct object

8. Which of the following is a transitive verb requiring a direct object?

(A) Precipitate
(B) Synthesize
(C) Deteriorate
(D) Oscillate

9. In 'Centrifugation separates organelle fractions', 'separates' is a:

(A) Finite transitive verb
(B) Gerund
(C) Non-finite verb
(D) Auxiliary verb

10. Which of the following contains an abstract noun?

(A) Sterile petri dish
(B) High enzymatic affinity
(C) Glass test tube
(D) Bacterial colony

11. Identify the coordinating conjunction: 'The broth was cloudy, but no odor arose.'

(A) was
(B) cloudy
(C) but
(D) arose

12. Which part of speech indicates spatial, temporal, or logical relationships?

(A) Interjection
(B) Preposition
(C) Adverb
(D) Determiner

13. In 'Incubate the culture at 37 deg C', the word 'at' is a:

(A) Preposition of time
(B) Preposition of condition
(C) Preposition of place/state
(D) Subordinating conjunction

14. Identify the gerund in: 'Pipetting viscous buffers requires calibrated tips.'

(A) Pipetting
(B) viscous
(C) requires
(D) calibrated

15. A word functioning as a linking (copular) verb in academic English is:

(A) Synthesize
(B) Remain
(C) Centrifuge
(D) Titrate

16. Which pronoun is classified as an indefinite pronoun?

(A) Which
(B) Those
(C) Everyone
(D) Whom

17. In 'The supernatant became turbid', 'turbid' functions as a:

(A) Predicate adjective
(B) Attributive adjective
(C) Manner adverb
(D) Direct object

18. Which form is the correct comparative degree for the adjective 'pure'?

(A) More pure
(B) Purer
(C) Most pure
(D) Purest

19. Which word in 'The enzyme reacted quite rapidly' is an adverb of degree?

(A) enzyme
(B) reacted
(C) quite
(D) rapidly

20. The acronym FANBOYS helps students remember which grammatical class?

(A) Correlative conjunctions
(B) Coordinating conjunctions
(C) Modal auxiliaries
(D) Relative adverbs

21. What type of pronoun is 'itself' in: 'The plasmid replicates itself'?

(A) Demonstrative
(B) Reflexive
(C) Interrogative
(D) Relative

22. Identify the collective noun in: 'The scientific panel approved the protocol.'

(A) scientific
(B) panel
(C) approved
(D) protocol

23. In the phrase 'two sterile flasks', the word 'sterile' is an:

(A) Attributive adjective
(B) Adverb
(C) Abstract noun
(D) Appositive noun

24. Which preposition correctly completes: 'The cell wall is composed ___ peptidoglycan'?

(A) with
(B) from
(C) of
(D) by

25. Which part of speech connects a subordinate clause to an independent clause?

(A) Coordinating conjunction
(B) Subordinating conjunction
(C) Preposition
(D) Interjection

26. In 'Although the primer annealed, no amplicon formed', 'Although' is a:

(A) Subordinating conjunction
(B) Coordinating conjunction
(C) Relative pronoun
(D) Conjunctive adverb

27. Which of the following words is a demonstrative pronoun?

(A) These
(B) They
(C) Their
(D) Them

28. Identify the infinitive form of the verb in technical writing:

(A) Transcribing
(B) Transcribed
(C) To transcribe
(D) Transcribes

29. Which word functions as a modal auxiliary indicating necessity?

(A) Might
(B) Could
(C) Must
(D) Would

30. In 'The sample was extremely volatile', 'extremely' modifies:

(A) The noun sample
(B) The linking verb was
(C) The adjective volatile
(D) The whole clause

31. Which sentence contains a reciprocal pronoun?

(A) Scientists respect one another.
(B) They isolated each gene.
(C) The cells divided themselves.
(D) Both samples were clear.

32. What part of speech is 'since' in: 'Since the pH dropped, the protein denatured'?

(A) Preposition of time
(B) Subordinating conjunction
(C) Adverb of degree
(D) Coordinating conjunction

33. Identify the proper noun among the following terms:

(A) Polymerase
(B) Escherichia coli
(C) Electrophoresis
(D) Spectrophotometer

34. Which of the following is a non-finite verb form?

(A) Amplifies
(B) Denatured (past tense)
(C) Denaturing (participle)
(D) Synthesized (past tense)

35. In 'Neither the sensor nor the pump operated', 'Neither...nor' is a:

(A) Correlative conjunction
(B) Coordinating conjunction
(C) Subordinating conjunction
(D) Compound preposition

36. Which of the following is a possessive pronoun that never precedes a noun?

(A) My
(B) Their
(C) Ours
(D) Your

37. Identify the preposition in: 'Transfer 5 mL into the sterile conical tube.'

(A) Transfer
(B) into
(C) sterile
(D) tube

38. What part of speech is 'rapidly' in: 'The lysate cleared rapidly'?

(A) Adjective
(B) Adverb of manner
(C) Conjunctive adverb
(D) Participle

39. In scientific registers, 'data' is historically derived as the plural form of:

(A) Datum
(B) Date
(C) Daton
(D) Datis

40. Which sentence uses an intransitive verb correctly?

(A) The pellet dissolved.
(B) The technician dissolved.
(C) The autoclave sterilized.
(D) The researcher purified.

41. Select the sentence with the correct scientific plural noun form:

(A) The bacteriums were killed.
(B) The criteria were satisfied.
(C) The medium were autoclaved.
(D) The phenomenons occurred.

42. In formal scientific prose, which relative pronoun introduces restrictive clauses?

(A) Which (with comma)
(B) That (without comma)
(C) Whom
(D) Whose

43. Identify the sentence where 'equipment' is used grammatically:

(A) New equipments were purchased.
(B) Several equipment was bought.
(C) New pieces of equipment arrived.
(D) Equipments are sterile.

44. Choose the correct preposition: 'The enzyme exhibits high specificity ___ galactose.'

(A) with
(B) towards / for
(C) against
(D) into

45. Which sentence contains nominalization creating an objective academic tone?

(A) We isolated the gene quickly.
(B) Isolation of the gene was rapid.
(C) I decided to purify the broth.
(D) They tested the lysate.

46. Identify the correct pronoun: 'Between you and ___, the calibration is invalid.'

(A) I
(B) me
(C) myself
(D) he

47. Which sentence demonstrates correct subject-verb concord with an indefinite pronoun?

(A) Each of the assays have failed.
(B) Each of the assays has failed.
(C) Each assays have failed.
(D) Each were positive.

48. Choose the correct compound adjective format preceding a noun:

(A) a high throughput screening
(B) a high-throughput screening
(C) a high throughput-screening
(D) a highly-throughput screening

49. In 'The cells, having reached confluence, were harvested', 'having reached' is a:

(A) Present gerund
(B) Perfect participle
(C) Past finite verb
(D) Bare infinitive

50. Which sentence uses the preposition 'between' correctly in laboratory reports?

(A) Divide the lysate between 4 tubes.
(B) Choose between the 2 protocols.
(C) Distribute between all flasks.
(D) Spread between 5 dishes.

51. Identify the conjunctive adverb connecting these two scientific clauses:

(A) The broth boiled; consequently, cells died.
(B) boiled
(C) consequently
(D) broth

52. Select the sentence where 'only' is positioned without ambiguity:

(A) The kit only detects DNA.
(B) The kit detects only DNA.
(C) Only the kit detects DNA.
(D) The kit detects DNA only after.

53. Choose the correct verb: 'The scientist ___ the sterile petri dish on the rack.'

(A) lay
(B) laid
(C) lain
(D) lied

54. In 'The turbidity of the medium decreased', what part of speech is 'turbidity'?

(A) Abstract noun
(B) Proper noun
(C) Collective noun
(D) Adjective

55. Which correlative conjunction pair correctly maintains parallel structure?

(A) both by PCR and sequencing
(B) both by PCR and by sequencing
(C) both PCR or sequencing
(D) not only PCR but sequencing

56. Identify the function of 'Since' in: 'Since 2020, CRISPR has revolutionized gene editing.'

(A) Subordinating conjunction
(B) Preposition of time
(C) Adverb of cause
(D) Relative pronoun

57. Select the sentence with correct capitalization and italicization of proper noun taxon:

(A) escherichia Coli was tested.
(B) Escherichia coli was tested.
(C) Escherichia Coli was tested.
(D) escherichia coli was tested.

58. Which preposition is required after the adjective 'deficient'?

(A) at
(B) with
(C) in
(D) from

59. In 'The yield increased significantly', 'significantly' is an adverb modifying:

(A) The noun yield
(B) The intransitive verb increased
(C) The subject article
(D) The object

60. Choose the correct pronoun: 'The committee published ___ consensus report.'

(A) their
(B) its
(C) it's
(D) theirs

61. Identify the sentence with an erroneous double comparative:

(A) The affinity was much higher.
(B) The yield was more greater.
(C) The reaction was far slower.
(D) The pH was slightly lower.

62. Which sentence uses an adverb of degree modifying another adverb?

(A) The cells grew remarkably well.
(B) The cells grew well.
(C) The cells grew quickly.
(D) The cells grew uniformly.

63. Select the sentence containing a dangling participle modifier:

(A) Staining the slide, the cells died.
(B) Staining the slide, I saw cells.
(C) The slide was stained by us.
(D) After staining, cells appeared.

64. Choose the correct preposition: 'The mutation resulted ___ non-functional protein.'

(A) to
(B) in
(C) from
(D) at

65. In 'Centrifuging at 14,000 rpm pelleted the debris', 'Centrifuging' functions as a:

(A) Noun subject (gerund)
(B) Verbal adjective (participle)
(C) Main finite predicate verb
(D) Prepositional object

66. Which sentence correctly uses 'fewer' versus 'less' in bio-measurements?

(A) Less colonies grew on the plate.
(B) Fewer colonies grew on the plate.
(C) Fewer biomass was collected.
(D) Fewer absorbance was detected.

67. Identify the sentence with the correct hyphenation of an adverb-adjective modifier:

(A) A chemically synthesized oligo
(B) A chemically-synthesized oligo
(C) A chemical-synthesized oligo
(D) A chemically synthesized-oligo

68. What part of speech is 'well' in: 'The recombinant protein folded well'?

(A) Adjective
(B) Adverb of manner
(C) Noun
(D) Interjection

69. Which subordinating conjunction introduces an antithetical contrast?

(A) Whereas
(B) Because
(C) Since
(D) Unless

70. In 'The investigator whom we interviewed published the paper', 'whom' is:

(A) Subjective relative pronoun
(B) Objective relative pronoun
(C) Possessive pronoun
(D) Demonstrative pronoun

71. Select the correct plural form of the singular scientific noun 'hypothesis':

(A) Hypothesis
(B) Hypothesises
(C) Hypotheses
(D) Hypothese

72. Identify the linking verb followed by a predicate adjective:

(A) The buffer smelled pungent.
(B) The buffer smelled pungently.
(C) The buffer mixed well.
(D) The buffer flowed quickly.

73. Choose the preposition: 'The protocol complied strictly ___ biosafety regulations.'

(A) with
(B) to
(C) for
(D) in

74. In 'None of the biomass was lost', 'None' takes a singular verb because:

(A) Biomass is an uncountable noun
(B) None is always singular
(C) Biomass is plural
(D) The sentence is passive

75. Which sentence uses the demonstrative pronoun 'that' for scientific comparison?

(A) Taq fidelity is higher than Pfu.
(B) Taq fidelity is higher than of Pfu.
(C) Taq fidelity is higher than that of Pfu.
(D) Taq fidelity is higher than those Pfu.

76. Identify the sentence that violates formal academic register by using a phrasal verb:

(A) The team investigated the mutation.
(B) The team looked into the mutation.
(C) The team analyzed the mutation.
(D) The team examined the mutation.

77. Select the sentence with correct agreement when compound subjects use 'neither...nor':

(A) Neither RNA nor proteins is intact.
(B) Neither RNA nor proteins are intact.
(C) Neither proteins nor RNA are intact.
(D) Neither were intact in the tube.

78. Which sentence correctly differentiates 'who' from 'whom' in research authorship?

(A) The scholar whom designed the primer...
(B) The scholar who designed the primer...
(C) The scholar which designed the primer...
(D) The scholar whom was awarded...

79. Identify the grammatical error in: 'The laboratory has three distinct apparatuses for PCR.'

(A) Apparatus cannot take plural -es
(B) Apparatus has no error here
(C) Distinct is redundant
(D) For should be with

80. Choose the sentence that correctly distinguishes 'affect' (verb) from 'effect' (noun):

(A) Temperature effects the rate.
(B) Temperature affects the rate.
(C) The mutate had a major affect.
(D) The drug effected no change.

81. In 'Data ___ collected over a three-month clinical observation period', the standard academic verb is:

(A) was
(B) were
(C) is
(D) has been

82. Identify the dangling modifier among the following options:

(A) Upon adding the substrate, color formed.
(B) Upon adding substrate, we saw color.
(C) When we added substrate, color formed.
(D) After substrate addition, dye reacted.

83. Select the sentence with a misplaced limiting adverb:

(A) The assay only revealed three mutant colonies.
(B) The assay revealed only three mutant colonies.
(C) Only the assay revealed colonies.
(D) Three mutant colonies only were revealed.

84. Choose the correct prepositional collocation: 'The bacteria were susceptible ___ ampicillin.'

(A) with
(B) to
(C) for
(D) towards

85. In 'The gene expression profile was different ___ our initial hypothesis', formal style requires:

(A) from
(B) than
(C) to
(D) with

86. Identify the sentence with an ambiguous pronoun reference:

(A) When the virus infected the cell, it lysed.
(B) The virus lysed the cell upon entry.
(C) Cellular lysis occurred post-infection.
(D) After viral infection, cells lysed.

87. Which sentence uses an unhyphenated compound noun vs a hyphenated compound adjective correctly?

(A) The flow rate was steady in the flow-cell.
(B) The flow-rate was steady in the flow cell.
(C) The flow-rate was steady in the flow-cell.
(D) The flow rate was steady in flow cell.

88. Select the sentence with faulty correlative conjunction parallelism:

(A) The drug is not only effective but safe.
(B) The drug not only kills but inhibits cells.
(C) The drug is not only potent but safe too.
(D) The drug not only acted but cured.

89. What is the grammatical error in: 'The research team consists out of six postdoctoral fellows'?

(A) Consists requires 'of', not 'out of'
(B) Team must be plural
(C) Out of is a proper compound preposition
(D) Six must be written in numerals

90. In 'The enzyme, as well as its cofactors, ___ required for catalytic turnover', the verb is:

(A) are
(B) is
(C) were
(D) have been

91. Identify the word that functions as a flat adverb in modern scientific English:

(A) Quickly
(B) Fast
(C) Accurately
(D) Precisely

92. Choose the correct relative pronoun: 'The biosensor, ___ electrodes were gold-plated, worked well.'

(A) which
(B) whose
(C) that
(D) whom

93. Select the sentence containing an incorrect preposition following 'concur':

(A) The PI concurred with the finding.
(B) The PI concurred in the decision.
(C) The PI concurred to the hypothesis.
(D) The PI concurred on the protocol.

94. In 'Quantifying gene copy numbers ___ crucial for viral load assessment', the correct verb is:

(A) are
(B) is
(C) were
(D) have been

95. Identify the sentence where 'as' functions as a preposition rather than a conjunction:

(A) The buffer served as a stabilizer.
(B) As the temperature rose, DNA melted.
(C) The gel ran as expected.
(D) As noted previously, Taq is stable.

96. Choose the correct pronoun: 'Every student submitted ___ thesis on microfluidic chips.'

(A) their
(B) his or her
(C) its
(D) theirs

97. Identify the sentence with correct placement of a split infinitive in formal style:

(A) To completely dissolve the pellet...
(B) To dissolve the pellet completely...
(C) Completely to dissolve the pellet...
(D) Dissolving completely the pellet...

98. In 'The number of viable cells ___ counted manually under inverted microscopy', the verb is:

(A) were
(B) was
(C) are
(D) have been

99. Select the sentence where 'due to' is used correctly as a predicate adjective complement:

(A) Due to the storm, the lab lost power.
(B) The power failure was due to the storm.
(C) Due to high heat, the enzyme denatured.
(D) Due to contamination, culture died.

100. Which sentence exhibits flawless syntactic agreement, prepositional collocation, and academic register?

(A) A series of assays was conducted to ascertain the dissociation constant.
(B) A series of assays were conducted to check out the dissociation constant.
(C) A series of assay was conducted to ascertain the dissociation constant.
(D) A series of assays was conducted for ascertaining the dissociation constant.

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Explain the structural taxonomy of the eight traditional parts of speech and discuss their specific syntactic functions in technical scientific prose.

Q2. Examine the grammatical distinction between countable and uncountable (mass) nouns in biotechnology, detailing common mass noun errors.

Q3. Discuss the role of nominalization in academic scientific discourse, evaluating both its stylistic benefits and cognitive disadvantages.

Q4. Analyze the rules governing subject-verb agreement (concord) with collective nouns in American versus British scientific English.

Q5. Examine the grammatical rules governing personal, possessive, and reflexive pronouns in scientific manuscripts.

Q6. Detail the syntactic and stylistic distinctions between 'which' and 'that' in restrictive and non-restrictive relative clauses.

Q7. Discuss indefinite pronouns and their subject-verb concord rules, focusing on tricky indefinite quantifiers in laboratory contexts.

Q8. Distinguish finite from non-finite verbs, classifying gerunds, participles, and infinitives in technical protocols.

Q9. Explain the grammatical errors of dangling and misplaced participles in biotechnology papers, providing clear correction methods.

Q10. Analyze transitive, intransitive, and ergative verbs in experimental biology procedures.

Q11. Examine the syntax of linking (copular) verbs and explain the selection of predicate adjectives versus adverbs.

Q12. Detail the rules governing degrees of comparison in adjectives and highlight errors in technical data descriptions.

Q13. Examine compound adjectives, hyphenation rules, and their role in scientific terminology.

Q14. Discuss the taxonomy of adverbs and analyze the placement of limiting adverbs like 'only', 'merely', and 'almost'.

Q15. Examine conjunctive adverbs in technical scientific transitions, punctuation requirements, and semicolon usage.

Q16. Analyze prepositional collocations and idiomatic verb-preposition pairings in molecular biology and biochemistry.

Q17. Examine the distinction between 'between' and 'among' in scientific data distribution and comparative analysis.

Q18. Analyze coordinating conjunctions (FANBOYS) and the rules for compound sentences versus compound predicates.

Q19. Discuss correlative conjunctions and the imperative of syntactic parallelism in academic biotechnology writing.

Q20. Examine subordinating conjunctions of cause, condition, and concession in experimental results reporting.

Q21. Analyze the grammatical status and capitalization rules for scientific binomial nomenclature and proper nouns.

Q22. Discuss flat adverbs versus -ly adverbs and common usage errors in technical laboratory reporting.

Q23. Examine the grammatical role of determiners and demonstrative adjectives in tracking scientific referents.

Q24. Detail the rules governing gender-neutral pronoun usage and avoidance of bias in contemporary biomedical writing.

Q25. Analyze prepositions of time: 'at', 'on', 'in', 'during', and 'for' in laboratory protocol documentation.

Q26. Examine the syntax and mechanics of the split infinitive in formal scientific and biomedical prose.

Q27. Discuss the proper syntactic deployment of Latin abbreviations (i.e., e.g., et al., etc., vs.) in academic publications.

Q28. Examine the grammatical rules governing 'fewer' versus 'less' and 'number' versus 'amount' in biological measurements.

Q29. Analyze the syntactic function of appositives and their punctuation in scientific manuscript writing.

Q30. Examine syntactic parallelism across bulleted technical lists and headings in biotechnology laboratory protocols.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1: [B]
Q2: [A]
Q3: [C]
Q4: [B]
Q5: [C]
Q6: [C]
Q7: [A]
Q8: [B]
Q9: [A]
Q10: [B]
Q11: [C]
Q12: [B]
Q13: [C]
Q14: [A]
Q15: [B]
Q16: [C]
Q17: [A]
Q18: [B]
Q19: [C]
Q20: [B]
Q21: [B]
Q22: [B]
Q23: [A]
Q24: [C]
Q25: [B]
Q26: [A]
Q27: [A]
Q28: [C]
Q29: [C]
Q30: [C]
Q31: [A]
Q32: [B]
Q33: [B]
Q34: [C]
Q35: [A]
Q36: [C]
Q37: [B]
Q38: [B]
Q39: [A]
Q40: [A]
Q41: [B]
Q42: [B]
Q43: [C]
Q44: [B]
Q45: [B]
Q46: [B]
Q47: [B]
Q48: [B]
Q49: [B]
Q50: [B]
Q51: [died]
Q52: [B]
Q53: [B]
Q54: [A]
Q55: [B]
Q56: [B]
Q57: [B]
Q58: [C]
Q59: [B]
Q60: [B]
Q61: [B]
Q62: [A]
Q63: [A]
Q64: [B]
Q65: [A]
Q66: [B]
Q67: [A]
Q68: [B]
Q69: [A]
Q70: [B]
Q71: [C]
Q72: [A]
Q73: [A]
Q74: [A]
Q75: [C]
Q76: [B]
Q77: [B]
Q78: [B]
Q79: [B]
Q80: [B]
Q81: [B]
Q82: [A]
Q83: [B]
Q84: [B]
Q85: [A]
Q86: [A]
Q87: [A]
Q88: [C]
Q89: [A]
Q90: [B]
Q91: [B]
Q92: [B]
Q93: [C]
Q94: [B]
Q95: [A]
Q96: [B]
Q97: [B]
Q98: [B]
Q99: [B]
Q100: [A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

[illegible]

analytical equipment".
- Hardware / Software: Strictly uncountable. Never write *hardwares or *softwares.
- Information / Knowledge: Mass nouns requiring partitive counters like "items of information".
- Media vs. Medium: 'Culture medium' is singular; 'culture media' is plural.
- Bacteria vs. Bacterium: 'Bacterium' is singular; 'bacteria' is plural.

3. Exemplary Scientific Corrections:
- Incorrect: "The lab ordered five new equipments and hardwares for genomic sequencing."
- Correct: "The laboratory procured five new pieces of equipment and computational hardware for genomic sequencing."
- Incorrect: "Different culture mediums was prepared for fungal sporulation."
- Correct: "Different culture media were prepared for fungal sporulation."
- Incorrect: "The data shows that this phenomenons is reproducible."
- Correct: "The data show that this phenomenon is reproducible."

[illegible]

- "The research team has published its findings."
- British English: Permits both singular and plural concord depending on semantic intention (notional concord). If individual members act discordantly, plural verbs are preferred:
- "The committee have disagreed on the allocation of research funds."

3. Contexts Requiring Plural Verbs:

When members of a collective unit perform individual, disparate actions simultaneously, plural concord is mandated in both styles:

- Correct: "The panel were divided in their evaluations of the clinical trial data."
- Stylistic Revision: "The members of the panel were divided in their evaluations..." (eliminates ambiguity).

[illegible]

- Relative Pronoun: 'Which' is strictly required.
- Example: "Taq polymerase, which was isolated from Thermus aquaticus, is thermostable."

3. Editing Diagnostic:
Test whether omitting the clause damages the logical identity of the subject noun phrase. If yes, use 'that' without commas; if no, use 'which' with commas.

[illegible] laboratory contexts.

adjectivally, or adverbially:
- "The assay was designed to quantify immunoglobulin titers." (Adverbial of purpose).

[illegible] papers, providing clear correction methods.

[illegible]

When two or more words combine to modify a following noun as a single unified concept, they are hyphenated to prevent syntactic ambiguity:
- Examples: "high-throughput screening", "double-stranded DNA", "temperature-sensitive mutant", "membrane-bound receptor", "site-directed mutagenesis".

2. The Post-Noun Predicative Exception:
When the compound adjective follows the noun (in predicate position), the hyphen is generally omitted:
- Pre-nominal: "We utilized a state-of-the-art flow cytometer."
- Post-nominal: "The flow cytometer was state of the art."

3. Adverb-Adjective Combinations Ending in -ly:
Adverbs ending in -ly never take a hyphen when preceding an adjective, because the -ly suffix already clearly signals an adverbial modifier:
- Incorrect: "a highly-conserved sequence", "a chemically-defined medium"
- Correct: "a highly conserved sequence", "a chemically defined medium"

[illegible]

completely eliminated."
- Sentence-Initial Position: Followed by a comma:
- Correct: "Consequently, the enzymatic reaction halted prematurely."
- Parenthetical (Mid-Clause) Position: Flanked by commas:
- Correct: "The initial hypothesis, therefore, requires substantial revision."

3. The Comma Splice Error:
Using a comma instead of a semicolon before a conjunctive adverb produces an ungrammatical comma splice:
- Incorrect: "The broth was cloudy, however, no bacteria grew."
- Correct: "The broth was cloudy; however, no bacteria grew."

[illegible]

- Condition: if, unless, provided that, assuming that.

2. Punctuation Convention:
- Initial Subordinate Clause: Must be followed by a comma:
- "Although the amplicon appeared faint, sequencing verified its identity."
- Final Subordinate Clause: Generally takes no comma if restrictive:
- "Sequencing verified its identity although the amplicon appeared faint."

[illegible]

Determiners introduce noun phrases and determine their referential specificity:
- Articles: Definite (the) and Indefinite (a, an).
- Demonstratives: this, that, these, those.
- Quantifiers: all, every, each, several, few, many, much.

2. Resolving Ambiguity with Demonstrative Adjectives:

A ubiquitous flaw in scientific papers is the 'unattended demonstrative pronoun'???using 'This' or 'That' as a solitary subject:
- Vague: "The enzyme precipitated at pH 4.0. This led to denaturation."
- Specific (Demonstrative Adjective + Summary Noun): "The enzyme precipitated at pH 4.0. This precipitation led to irreversible denaturation."
- By supplying the summary noun, the writer reinforces cognitive continuity and prevents referential ambiguity.

[illegible] biomedical writing.

- For: Expresses the total continuous duration of an experimental condition:
- "Incubate the suspension for 48 hours."
- During: Specifies that an event occurred within the timeframe of an ongoing procedure:
- "Contamination was detected during the second passage."

[illegible] academic publications.

[illegible]

- 2. Addition of 5 uL of template DNA. (Nominalization / Noun phrase)
- 3. You must vortex the tubes gently for 10 seconds. (Second-person pronoun + modal)
- 4. Centrifugation of the samples at 2,000 g. (Gerund / Noun phrase)

3. Fully Parallel Revision (All Imperative Verbs):
- 1. Pipette 50 uL of master mix into each reaction tube.
- 2. Add 5 uL of template DNA to the mixture.
- 3. Vortex the tubes gently for 10 seconds.
- 4. Centrifuge the samples at 2,000 g for 30 seconds.
- Result: Maximizes readability, eliminates cognitive drag, and prevents procedural operational errors in cleanroom environments.
